

Problem 2a)

a) The probability space is $(\Omega, \mathcal{F}, \mathbb{P})$:

i) Sample space (Ω) : All permutations of placing six cups on 6 fixed saucers. (as the order of the saucers is irrelevant, we can arbitrarily fix them). Therefore, we have $|\Omega| = 6! = 720$ possible outcomes; *There are only 90 outcomes since cups of the same color can't be told apart*
 $\Omega = \{ \omega := (\omega_1, \dots, \omega_6) \mid \omega \text{ is a permutation of } C \}, \text{ where } C = \{c_1, \dots, c_6\} \text{ is the set of cups}$

ii) Event space $(\mathcal{F})$: $\mathcal{F} = \mathcal{P}(\Omega)$ as Ω is finite

iii) Probability measure $(\mathbb{P})$: Since every outcome is equally likely:

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} \quad \text{where } E \subseteq \mathcal{F} \text{ is an event.}$$

Let E be the event that no cup is placed on a saucer of the same color. Then E^c is the event that at least one cup is placed on a saucer of the same color ($E^c = \Omega \setminus E$) and $\mathbb{P}(E) = 1 - \mathbb{P}(E^c)$. Let A_k be the event that a the cup c_k ($c_k \in C$) is placed in a saucer of the same color, so that $E^c = \bigcup_{k=1}^6 A_k$, and by the inclusion-exclusion criteria we have that
 $\mathbb{P}(E^c) = \mathbb{P}\left(\bigcup_{k=1}^6 A_k\right) = \sum_{i=1}^6 (-1)^{i-1} B_i$, where B_i is the sum of probabilities of all intersections of i A_k events.

The table below calculate the values for S_k *What does S_k stand for?*

S_k	<u>Result</u>	Type of permutation	Occurrences	Probability
S_1	$6 \cdot \frac{1}{3} = 2$	Single event	$\binom{6}{1} = 6$	$\binom{2}{1} 5! \cdot \frac{1}{6!} = \frac{1}{3}$
S_2	$3 \cdot \frac{1}{15} + 12 \cdot \frac{2}{15} = \frac{9}{5}$	Two cups of same color	$\binom{3}{1} \binom{3}{1} = 3$	$\binom{2}{1} \binom{1}{1} \cdot 4! \cdot \frac{1}{6!} = \frac{1}{15}$
		Two cups of different color	$\binom{3}{2} \cdot \binom{2}{1} \binom{2}{1} = 12$	$\binom{2}{1} \binom{2}{1} \cdot 4! \cdot \frac{1}{6!} = \frac{2}{15}$
S_3	$12 \cdot \frac{1}{30} + 8 \cdot \frac{1}{15} = \frac{14}{15}$	Two cups of same color and one different	$\left(\binom{3}{1} \binom{2}{1} \binom{2}{1}\right) \cdot \binom{2}{1} \binom{2}{1} = 12$	$\binom{2}{1} \binom{1}{1} \binom{2}{1} \cdot 3! \cdot \frac{1}{6!} = \frac{1}{30}$
		Three cups of different color	$\binom{3}{3} \binom{2}{1} \binom{2}{1} \binom{2}{1} = 8$	$\binom{2}{1} \binom{2}{1} \binom{2}{1} \cdot 3! \cdot \frac{1}{6!} = \frac{1}{15}$
S_4	$3 \cdot \frac{1}{90} + 12 \cdot \frac{1}{45} = \frac{3}{10}$	Two pairs of cups of the same color	$\binom{3}{2} \binom{2}{2} \binom{2}{2} = 3$	$\binom{2}{1} \binom{1}{1} \binom{2}{1} \binom{1}{1} 2! \cdot \frac{1}{6!} = \frac{1}{90}$
		One pair of the same color, one pair of different color	$\binom{3}{1} \binom{2}{2} \binom{2}{2} \binom{2}{1} \binom{2}{1} = 12$	$\binom{2}{1} \binom{1}{1} \binom{2}{1} \binom{2}{1} 2! \cdot \frac{1}{6!} = \frac{1}{45}$
S_5	$6 \cdot \frac{1}{90} = \frac{1}{15}$	Two pairs of the same and one cup of different	$\binom{3}{2} \binom{2}{2} \binom{2}{2} \cdot \binom{1}{1} \binom{2}{1} = 6$	$\binom{2}{1} \binom{1}{1} \binom{2}{1} \binom{1}{1} \binom{2}{1} 1! \cdot \frac{1}{6!} = \frac{1}{90}$
S_6	$1 \cdot \frac{1}{90} = \frac{1}{90}$	Three pairs of the same color	$\binom{3}{3} \binom{2}{2} \binom{2}{2} \binom{2}{2} = 1$	$\binom{2}{1} \binom{1}{1} \binom{2}{1} \binom{1}{1} \binom{2}{1} \binom{1}{1} 0! \cdot \frac{1}{6!} = \frac{1}{90}$

Therefore, $\mathbb{P}(E^c) = 2 - \frac{9}{5} + \frac{14}{15} - \frac{3}{10} + \frac{1}{15} - \frac{1}{90} = \frac{8}{9}$, and the

probability that no cup is placed on a saucer with the same

color is $\mathbb{P}(E) = 1 - \mathbb{P}(E^c) = 1 - \frac{8}{9} = \frac{1}{9}$

Idea is correct - But how you got to the result is unclear

b) The probability space of a single round of rolling two dice is $(\Omega_0, \mathcal{F}_0, \mathbb{P}_0)$:

- $\Omega_0 = \{(a, b) \mid a, b \in \{1, \dots, 6\}\}$ Ω_0 should contain the sum of a and b . The probabilities then aren't evenly distributed
- $\mathcal{F}_0 = \mathcal{P}(\Omega_0)$
- $\mathbb{P}_0(E_0) = \frac{|E_0|}{|\Omega_0|}$ $E_0 \in \mathcal{F}_0$

The full probability space is $\Omega = \Omega_0^{\mathbb{N}}$, $\mathcal{F} = \bigotimes_{i=1}^{\infty} \mathcal{F}_0$ and $\mathbb{P} = \bigotimes_{i=1}^{\infty} \mathbb{P}_0$

Let E_x be the event that the sum of the dice is x in a single round s.t.

$$\bullet E_6 = \{(5,1), (4,2), (3,3), (2,4), (1,5)\}, |E_6| = 5, \mathbb{P}(E_6) = \frac{5}{36}$$

$$\bullet E_7 = \{(6,1), (5,2), (4,3), (3,4), (2,5), (1,6)\}, |E_7| = 6, \mathbb{P}(E_7) = \frac{1}{6}$$

$$\text{Then: } \mathbb{P}(E_6^c) = 1 - \mathbb{P}(E_6) = \frac{31}{36} \text{ and } \mathbb{P}(E_6^c) = 1 - \mathbb{P}(E_6) = \frac{5}{6}$$

The overall game follows identical cycles of four games ABBA until a player wins (i.e., E_6 occurs in a round A, or E_7 in a round B). In a single cycle, A wins the cycle if A wins in turn 1 or 4 of the cycle.

Let T_{Pk} be the event that player $P \in \{A, B\}$ wins in round k of a single cycle: $\mathbb{P}(T_{P1}) = \mathbb{P}(E_A) = \frac{5}{36}$, $\mathbb{P}(T_{P2}) = \mathbb{P}(T_{P3}) = 0$ and

$$\mathbb{P}(T_{P4}) \stackrel{(\text{ind})}{=} \mathbb{P}(E_A^c) \mathbb{P}(E_B^c) \mathbb{P}(E_B^c) \mathbb{P}(E_A) = \frac{31}{36} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{36} = \frac{3875}{46656}. \text{ Then,}$$

let C_w be the event that player A wins a single cycle s.t

$$\mathbb{P}(C_w) = \sum_{k=1}^4 \mathbb{P}(T_{Ak}) = \frac{10355}{46656} \text{ and let } C_t \text{ be the event that there}$$

is no winner in a single cycle, s.t., $\mathbb{P}(C_t) \stackrel{(\text{ind})}{=} \mathbb{P}(E_A^c) \mathbb{P}(E_B^c) \mathbb{P}(E_B^c) \mathbb{P}(E_A^c)$

$$= \frac{31}{36} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{31}{36} = \frac{24025}{46656} \quad \checkmark$$

Let W_A be the event of winning the overall game, i.e.,

$$W_A = \bigcup_{j=1}^{\infty} C_{wj}, \text{ where } j \text{ indicates the } j\text{-th cycle of the game with}$$

$$\mathbb{P}(C_{wj}) \stackrel{(\text{ind})}{=} \mathbb{P}(C_t)^{j-1} \mathbb{P}(C_w). \text{ Therefore =}$$

$$\mathbb{P}(W_A) = \mathbb{P}\left(\bigcup_{j=1}^{\infty} C_{wj}\right) \stackrel{(\sigma\text{-additivity})}{=} \sum_{j=1}^{\infty} \mathbb{P}(C_t)^{j-1} \mathbb{P}(C_w) = \frac{\mathbb{P}(C_w)}{1 - \mathbb{P}(C_t)}$$

$$= \frac{\frac{10355}{46656}}{1 - \frac{24025}{46656}} = \frac{10355}{22631} \approx 0.458$$

The chances of winning for A are ≈ 0.458 . $\checkmark$

$$\frac{9.5}{6}$$

$$\frac{3}{3}$$